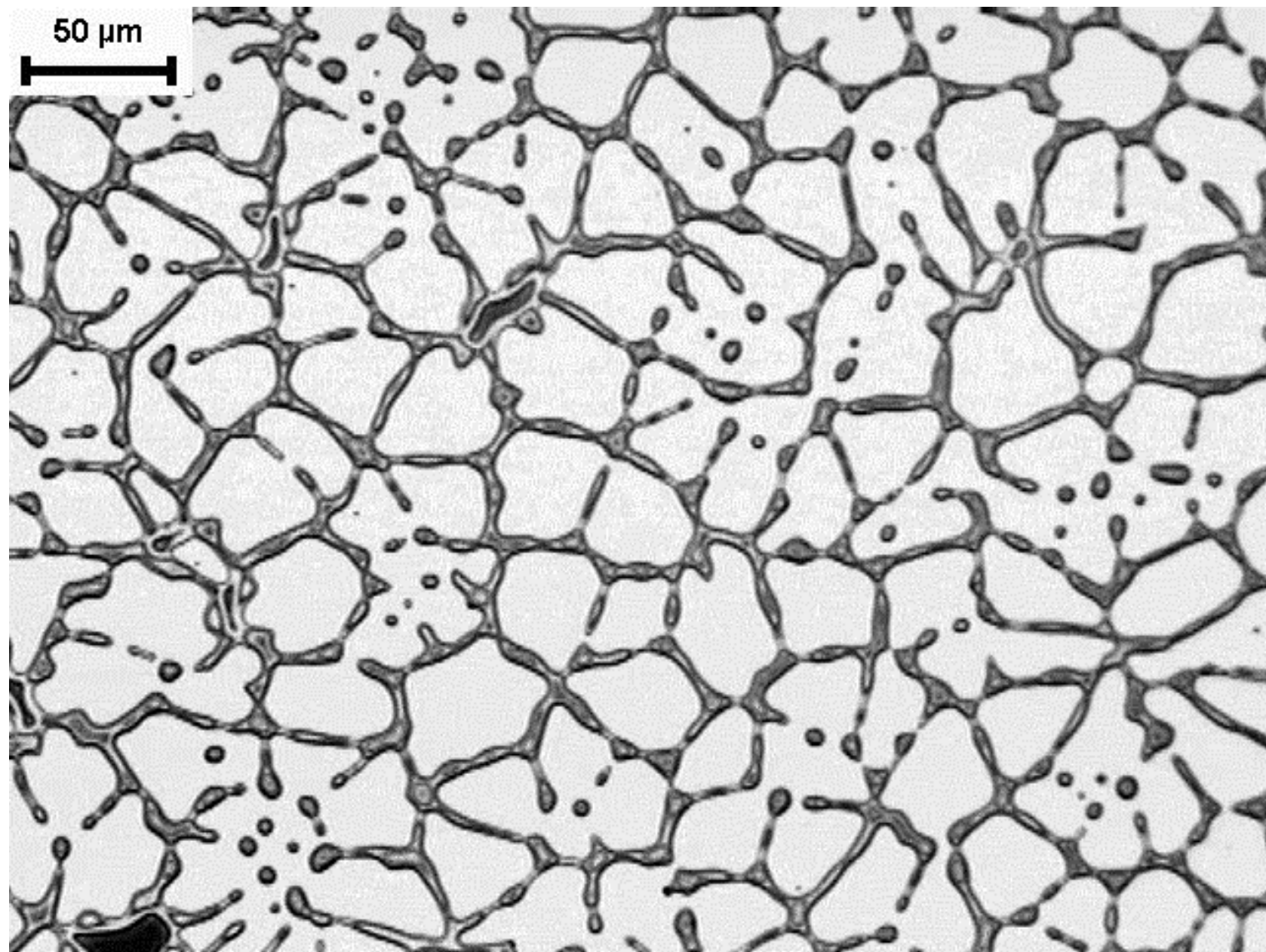


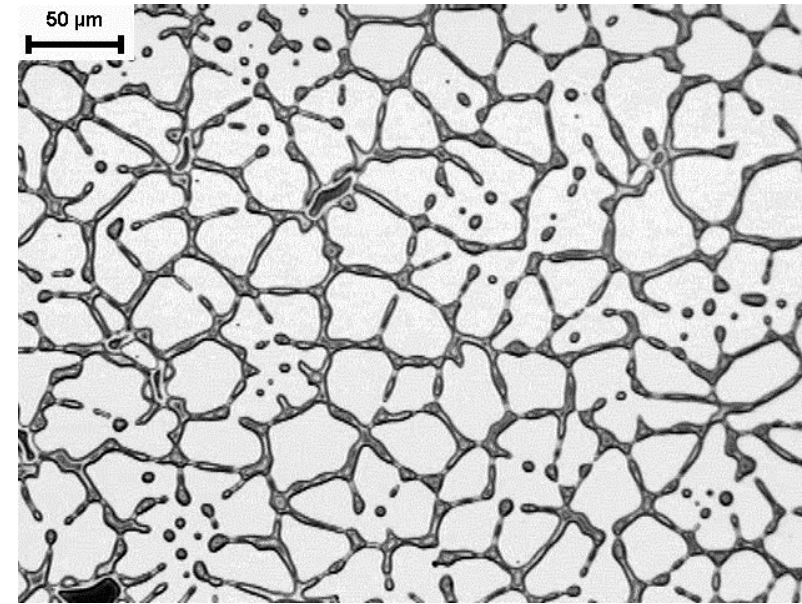
Spontaneous Instability in Ultra Thin Polymer Films

CH 62052: Instability

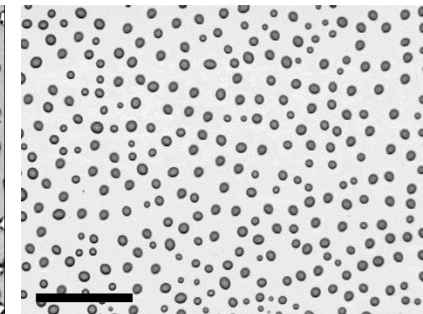
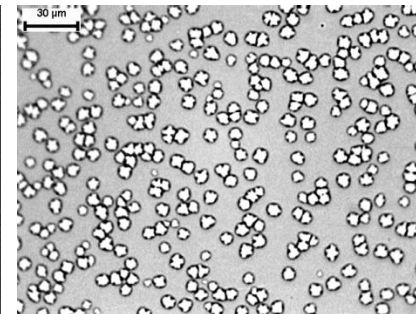
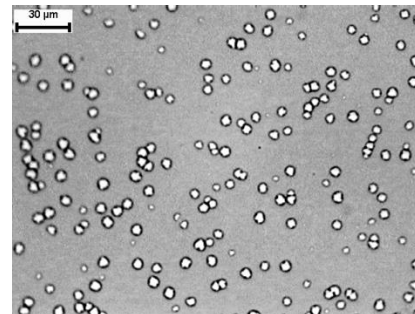
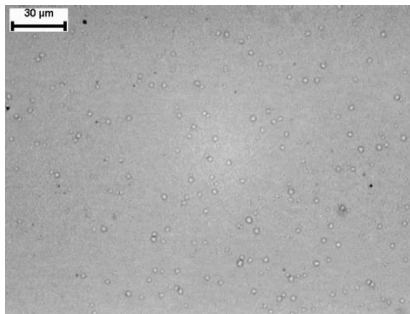




In-situ observation of Dewetting



Sequence: Growth of undulations, hole formation, rim formation, hole growth, cellular ribbon network, disintegration of the ribbons, droplet formation



Thin Film Instability and Polymer

The phenomena is observed in liquid films, as morphological evolution involves flow!

However, thin film dewetting experiments are generally not performed with regular liquids like water or organic solvents.

Such liquids will be prone to evaporation and thus the film thickness will continuously change.

It is here glassy polymer thin films become extremely handy, as they can be:

- 1) Taken to liquid or molten state.
- 2) Can be quenched to freeze the evolution.
- 3) Low vapour pressure so almost no evaporative loss
- 4) Solid final dewetted structures allow detailed high magnification microscopy
- 5) High viscosity polymers lead to slower dynamics and therefore allow very easy time resolved experiments.

